Graphene/2D Materials Based Field-Effect Devices and Optoelectronics: Graphene/Silicon Avalanche Photodetectors

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In this report we present a summary of the nanofabrication steps taken to realise graphene-silicon photodetectors, together with the results obtained using photocurrent mapping. The devices have planar circular geometry, which is expected to remove the corner effects present in the previously demonstrated devices with rectangular geometry. The fabrication was done on a silicon-on-insulator substrate, motivated by the advantages that it offers over its bulk silicon counterpart.

I. Introduction

The possibility of creating hybrid platforms with graphene and silicon is of paramount interest due to the fact that it allows the integration of graphene in complementary metal-oxide-semiconductor (CMOS) techonology [1]. Previously, graphene-silicon (GS) Schottky junction photodiodes were demonstrated to be an efficient platform for photodetection, with possible photovoltaic applications. However, the performance of such devices in terms of response speed remains below that of the commerical silicon-based photodetectors or that of the more common graphene-based photodetectors using metal-graphene-metal (MGM) architectures [2].

One solution proposed, as an improvement of the response speed of graphene-silicon Schottky juction devices, was to replace the commonly employed bulk silicon substrate with a silicon-on-insulator (SOI) one [3]. This substrate consists of a thin silicon layer on top of a buried silicon oxide layer (BOX), and it comes with several advantages over its bulk silicon counterpart. The most significant one is that it allows neighbouring devices to be fully isolated dielectrically. Also, SOI has a reduced capacitive coupling and less leakage currents, making it a suitable choice for low power and high speed applications.

The performance of these devices has been studied in two different scenarios, one employing a graphene-SOI Shottky diode with planar and rectangular geometry (GSOI-planar) and the other having a grating surface topography (GSOI-grating). It has been found that the operating speed for these devices is more than two orders of magnitude greater than that of the graphene-silicon devices based on bulk silicon [3].

Following similar nanofabrication steps as the ones used previously for the GSOI-planar Shottky diode, in our experiment we aim to demonstrate the realisation of such devices with circular geometry instead of a rectangular one. This is motivated by symmetry considerations, since any corner effects that might arise in a rectangular geometry will be avoided.

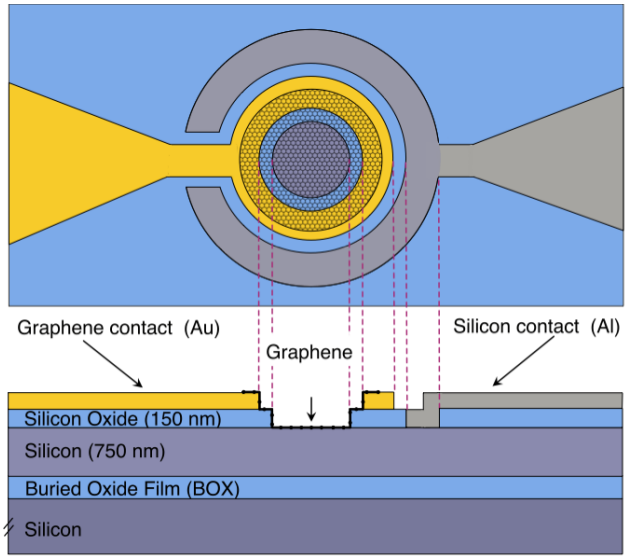
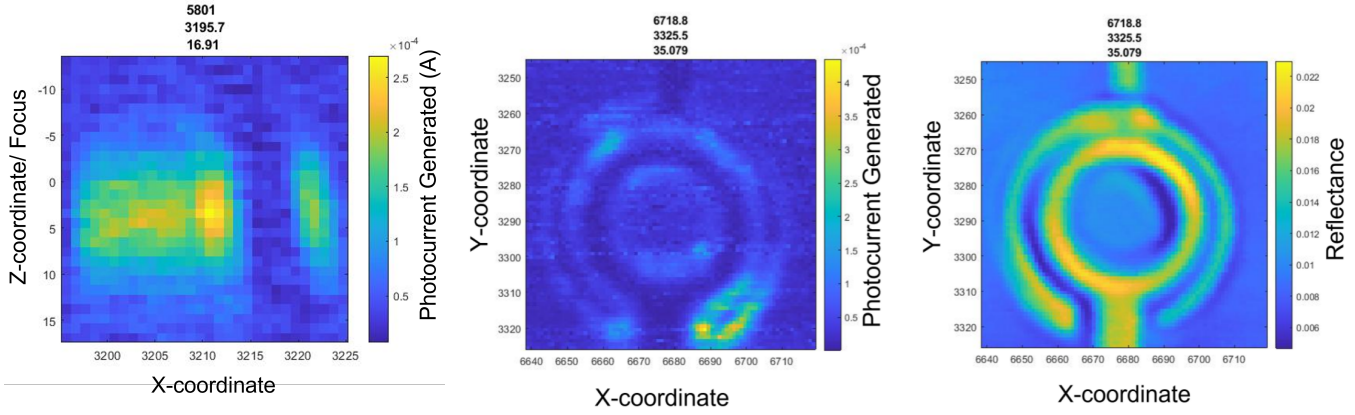


FIG. 1. Schematic illustration of one our devices: top-view in the upper part and side-view on the bottom, as seen by slicing through the middle section. The fact that part of the Al layer sits on the Si trench is emphasised by a slight difference in contrast. Although the shape of the contact pads is exact, their size is scaled down greatly in order to fit into the figure. Note that the smallest feature on all our devices, i.e. between adjacent rings or trenches, is 5 µm.

II. Fabrication Process

For the fabrication of this GSOI Schottky diode photodetector, we started with an n-doped SOI wafer with a Si thickness of approximately 10 µm and SiO_2 of 2-3 µm. We scribbed it and then KOH etched it for approximately 6 hours and 30 minutes in a solution of concentration 30%, i.e. 7.5 g KOH and 17.5 ml de-ionised (DI) water. This yielded a Si thickness of approximately 750 nm in the central part of the wafer. We fabricated a total of 16 devices (arranged as a 4x4 matrix), with every row having 4 identical copies and each column slightly different dimensions - inner graphene ring radii of 25 nm, 50 nm, 100 nm, and 150 nm. An example for the architecture of one of our devices is shown in Figure 1.



(a) X-Z Photocurrent Map (b) X-Y Photocurrent Map (c) X-Y Reflectance Map

FIG. 2. (a) Photocurrent map of device mapped in x and z coordinates illuminated by high power pulsed laser of wavelength 1,550 nm. Highest activity can be seen at the edges of the graphene ring and outside of the Au contact, implying this photocurrent is generated by Si rather than Graphene. (b) Photocurrent map and (c) reflectance map of device illuminated by lower power continuous laser of wavelength 1,550 nm.

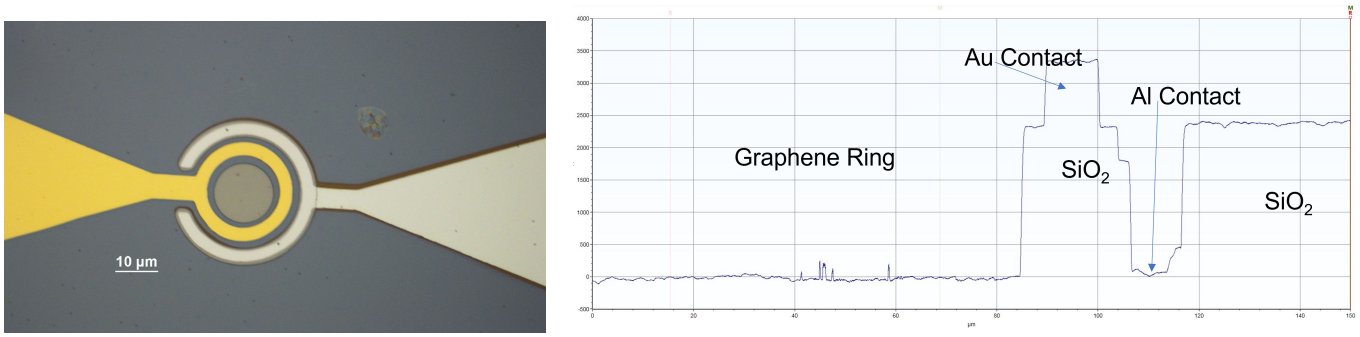
Firstly, we thermally grew a 150 nm SiO_2 layer using a furnace set at 1,050°C for about 35 minutes. Using S1805 as a photoresist, we spin coated it for 60 seconds at 4,000 rotations per minute (RPM) and baked it at 110°C for 3 minutes to obtain a 460 nm layer. Using the laser writer set at a power of 402 nJ cm^{-2}, gain 133 and D-step 4, we masked the sample with a pattern consisting of a circle surrounded by a ring with an open trench at one end, which is shown in purple in the top part of Figure 1. This was developed in MF319 for 1 minute and then gently agitated in DI water to make sure the exposed mask is properly cleaned. Then, we used a 6:1 Buffered Oxide Etch (BOE) solution with 49% HF concentration to remove the 150 nm of oxide, for which we calculated a rough etch time of 3.5 minutes. The excess mask was removed in acetone.

Next, we did a bilayer mask consisting of PMGI and S1805 in order to make the gold contact. We spin coated a 300 nm PMGI layer at 3,000 RPM for 60 seconds and baked it at 160°C for 10 minutes, while for the S1805 mask we used the same recipe as before. For this lithography step we patterned a contact pad shaped as an irregular pentagon with two parallel sides, attached to a ring between the Si inner circle and the Si trench we etched before - which is shown in yellow in Figure 1. Au was then deposited in the exposed pattern using thermal evaporation, and the residual mask was removed using a bilayer lift-off in acetone.

The Aluminium contact had a similar pattern: a contact pad identical to the Au one, attached to an open ring perfectly aligned on top of the Si ring etched before. This is coloured in gray in Figure 1, while the region where the Si and Al are on top of each other has a slightly purple shade. Different from the Au contact for which we did a bilayer, for the Al mask we instead used a single layer of S1813 photoresist. Although a bilayer lift-off might have been better, it was not feasible for our device as MF319 (which is the developer for the S1805 photoresist) dissolves Al. Choosing instead S1813 was motivated by the fact that it develops in acetone and it also gives a much thicker mask than S1805 (by almost three times), which was needed for a single layer. Right before the thermal evaporation, the sample was dipped into a BOE solution for 30 seconds, in order to remove the native oxide.

We then used graphene grown by chemical vapour deposition (CVD) on copper foil, which we cut into a rectangular shape of similar size with our wafer's active area and attached it on a dummy Si wafer using Kapton tape. We spin coated it with a 200 nm thick layer of poly-methyl-methacrylate (PMMA) at 2,000 RPM for 60 seconds and baked it for 3 minutes on the hotplate at 130°C. This step was needed to protect the graphene on the top side of the Cu foil. We then flipped it on the other side and O_2/Ar plasma etched it for 60 seconds in the Moorfield NanoETCH, in order to remove the graphene layer grown on the other side of the foil. Next, we left the graphene/PMMA membrane to etch in an ammonium persulfate (APS) solution of concentration 2% (i.e. 0.5 g of APS in 25 ml DI water) overnight.

Next day, using a dummy Si 90 nm wafer we transferred the graphene/PMMA membrane from the APS solution into DI water, where we left it for 30 minutes to rinse. In order to make sure we properly remove both the etchant and Cu residues, we transferred it again into clean DI water for another 20 minutes. This was repeated once again. From there we fished it on our sample, dried it at an angle, left it on the hot plate for 3 minutes at 130°C and cleaned it: acetone for 5 minutes, another fresh acetone - 10 minutes, isopropyl alcohol solution (IPA) - 5 minutes and hexane - 3 minutes.



(a) Optical Image

(b) DekTak

FIG. 3. (a) Optical microscopy image of our device taken with a magnification of 50, before the graphene etching step. The scale of the image is 10 µm. (b) DekTak measurement of a broken device taken after the contacts were deposited. In this subfigure the scale of the y-axis is Angstroms (Å), starting from -500 as the lower limit and going in increments of 500 up to 4000. The scale of the x-axis is micrometers (µm), starting from 0 and going up to 150 in increments of 20.

A microscope image of one of our devices can be seen in Figure 3(a), where the tiny black dots are characteristic to CVD graphene.

Lastly, we did an additional lithography step to remove the graphene outside our devices which might short-circuit them. The pattern we wanted for graphene was a disk on top of the ring-like Au contact, which is shown in black in Figure 1. In order to protect the Al from being dissolved by the MF319 developer, we chose to also include the Al contact in this mask. For this, we spin coated S1805 on our sample, did laser writing on the inverted pattern for both graphene and Al (i.e. exposing everything but the graphene ring and the Al contact), and O_2/Ar plasma etched the excess graphene for 60 seconds in the Moorfield NanoETCH. Finally, we cleaned the remaining photoresist in acetone, IPA and hexane.

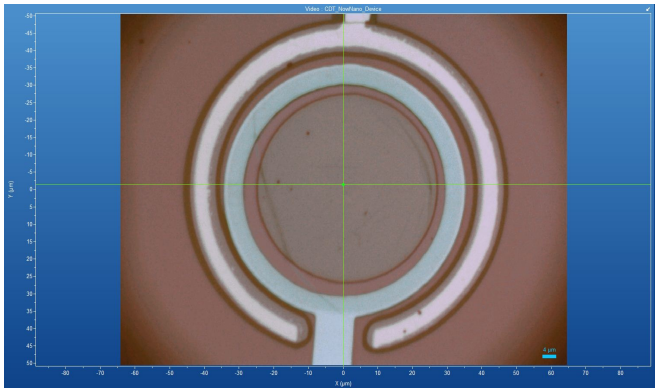
III. Results and Discussion

Two of our devices were tested without illumination, in ambient conditions and showed rectifying behaviour, similar to a diode. The device also showed breakdown behaviour around 13 V, which is a lower value than predicted in the simulations. There are a number of factors that could have led to this, such as decreased conductivity of graphene layer due to polymer residue or less direct contact area between the graphene and Si for current transfer.

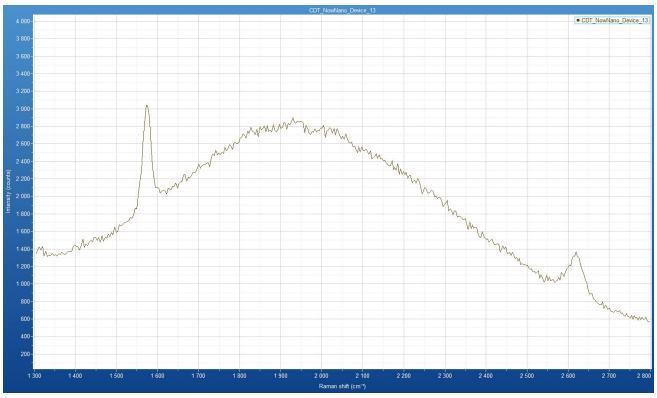
Under general illumination there was an increase in the detected current, showing that the device does operate as a photodetector. To investigate the contribution of graphene to the photocurrent, the device was illuminated by a laser of wavelength 1,310 nm, above the Si photo-active region of spectrum, and a very small photocurrent of 80 pA was detected. Mapping of the photocurrent of the device revealed that this photocurrent was not produced by the graphene, but rather by 2-photon absorption of Si. This is shown in Figure 2(a), which shows that photocurrent was produced in exposed Si regions at the edge of the graphene circle, and crucially outside of the Au contact, where no graphene in present. The mapping was also conducted with variations in the Z coordinate/focus of the laser, and the photocurrent detected is independent of focus, implying that the photocurrent is being produced in the underlying Si rather than the graphene layer.

X-Y photocurrent mapping was conducted on a different device as well as reflectance measurements, which can be seen in Figure 2(b) and Figure 2(c), respectively. In this device the highest photocurrent was produced by a region around the Al contact ring which appears to have less Al deposited compared to the rest of the ring. This conclusion is supported by the reflectance map, and also by Dektak measurements of the devices taken after the contacts were deposited, as seen in Figure 3(b). From this plot it is clear that the Al contact is very thin ($\approx$ 10 nm), meaning that there might be regions where the exposed Si is not entirely covered by Al. The photocurrent generated in this region is of the same order as the photocurrent generated at the edge of the graphene ring in Figure 2(a), and so it is likely that some graphene, which was not etched from the Al contact ring, is directly contacting Si and generating photocurrent.

Another possible reason for the lack of photocurrent generated by the central graphene ring is that the graphene was of relatively low quality. This was evaluated by conducting Raman spectroscopy on all of the devices after transferring graphene onto the Si wafer. The majority of the devices showed the characteristic G and 2D peaks of graphene, but had considerable variations from the Raman plot of pristine graphene. Many of the devices had residual polymer still on the graphene, as can be seen in Figure 4(a) after transfer due to the fact that a single layer photoresist was used to protect the graphene from O_2/Ar plasma etching.



(a) Screenshot of Optical Image



(b) Raman Spectrum

FIG. 4. (a) Optical microscopy image of one device with (b) associated Raman spectrum. Graphene characteristic peaks are present but there is a large peak between the G and 2D peaks that is believed to be due to polymeric residue. 2D:G peak intensity ratio is low, implying a high concentration of defects in the graphene. The scale in Figure (a) is 4 µm.

This resulted in an incomplete mask removal compared to the ideal case in which a double layer mask would had been used. However, this was unavoidable, as using a bilayer resist with PMGI would have removed the Al contact when the resist was developed. This residual polymer would have lowered the electrical activity of the graphene layer and so may have reduced photocurrent generation. It can also be seen in Figure 4(b) that the 2D:G peak intensity is low, which implies that the graphene present has a high concentration of defects. This would decrease the electrical properties of the graphene and could also inhibit the generation of photocurrent.

IV. Conclusions

Whilst our devices did not perform as expected, they did show rectifying behaviour and generated some photocurrent under illumination, showing that we had successfully manufactured a photodetector. In addition, due to the photocurrent generation of Si underneath the graphene, we can say that we have made a transparent diode electrode which could have potential applications in future technologies. The confirmation that this considered architecture produces a functioning photodetector is encouraging for successive devices utilising similar designs to these, and so there is great potential that these may allow for the realisation of graphene/Si avalanche photo detectors in the near future.

Acknowledgments

We would like to thank you again for assigning us this amazing project. We surely learnt a lot from it and we are grateful that it involved so many different experimental techniques that we could not have covered in only 6 weeks otherwise. We are in particular thankful to Will for all the time and the extra effort that he put into helping us further this lab project. Sorry for all those early mornings and late evenings that we put you through! Lastly, we would also like to thank Nawal and Patrick for helping with the characterization of the devices in the last week.

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[2] T. Mueller, F. Xia, and P. Avouris, Graphene photodetectors for high-speed optical communications, Nature photonics **4**, 297 (2010).

[3] H. Selvi, E. W. Hill, P. Parkinson, and T. J. Echtermeyer, Graphene–silicon-on-insulator (gsoi) schottky diode photodetectors, Nanoscale **10**, 18926 (2018).